Introduction to basic networking concepts: TCP, UDP, IP

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Introduction to Basic Networking Concepts

Networking is the backbone of modern communication, enabling devices, applications, and users to share data efficiently. Three fundamental protocols in networking are:

- **TCP (Transmission Control Protocol)** – Ensures reliable communication between devices.
- **UDP (User Datagram Protocol)** – Provides fast but unreliable data transmission.
- **IP (Internet Protocol)** – Handles addressing and routing of data across networks.

These protocols form the foundation of the **Internet Protocol (IP) suite**, which enables network communication worldwide. Understanding these protocols is essential for cybersecurity, network management, and ethical hacking.

Internet Protocol (IP) – The Addressing System

What is IP?

IP (Internet Protocol) is the fundamental protocol responsible for identifying, addressing, and routing packets across networks. It provides **unique addresses** to devices and ensures that data reaches the correct destination.

Key Functions of IP:

🔹 **Addressing:** Assigns a unique IP address to each device in a network.

🔹 **Packet Routing:** Directs data packets to their destination based on IP addresses.

🔹 **Fragmentation & Reassembly:** Splits large data packets into smaller ones for transmission and reassembles them at the receiver's end.

Types of IP Addresses:

- ❖ **IPv4 (Internet Protocol Version 4)** – Uses a **32-bit address** (e.g., `192.168.1.1`), allowing **4.3 billion unique addresses**.
- ❖ **IPv6 (Internet Protocol Version 6)** – Uses a **128-bit address** (e.g., `2001:db8::ff00:42:8329`), supporting **340 undecillion addresses**.

IP Address Classification (IPv4):

Class	Address Range	Default Subnet mask	Usage
Class A	1.0.0.0 – 126.255.255.255	255.0.0.0	Large networks (e.g., ISPs, big enterprises)

Class B	128.0.0.0 – 191.255.255.255	255.255.0.0	Medium-sized networks
Class C	192.0.0.0 – 223.255.255.255	255.255.255.0	Small networks (e.g., home networks)
Class D	224.0.0.0 – 239.255.255.255	N/A	Multicasting
Class E	240.0.0.0 – 255.255.255.255	N/A	Experimental

IP Packet Structure:

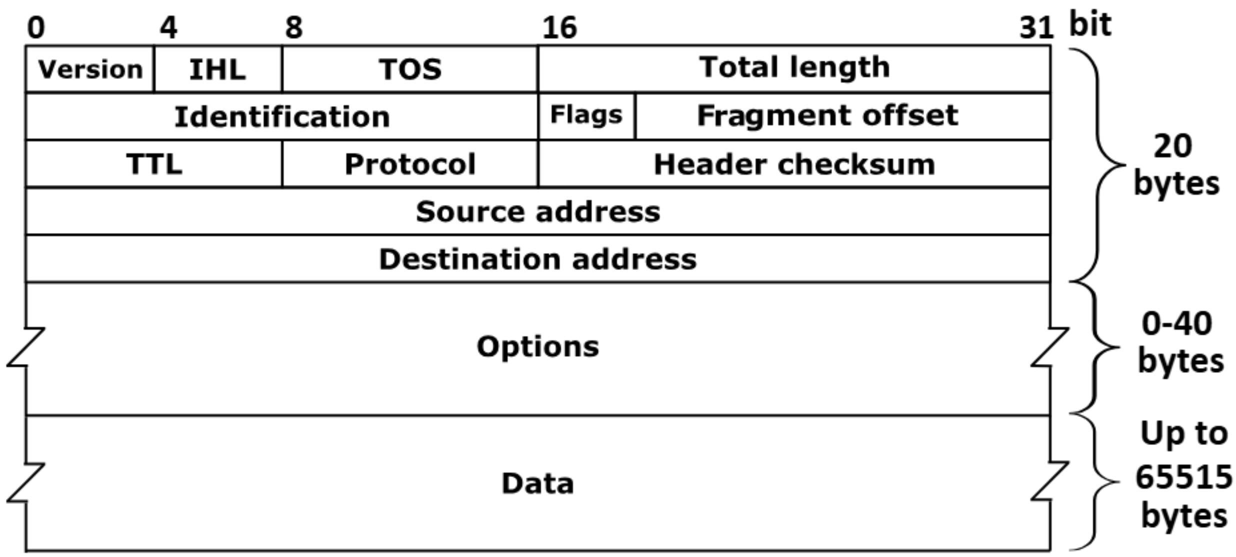
Each IP packet consists of **two main parts:**1️⃣**Header** – Contains metadata such as source/destination IP, time-to-live (TTL), checksum, etc.

2️⃣**Payload** – Contains the actual data being transmitted.

IP Header Fields (IPv4):

Field	Size (Bits)	Description
Version	4	Specifies IPv4 or IPv6
Header Length	4	Size of the header
Type of Service (Tos)	8	Defines priority & QoS
Total Length	16	Total size of packet
Identification	16	Unique ID for packet fragmentation
Flags	3	Controls fragmentation
Fragment Offset	13	Indicates fragment position

Time-to-Live (TTL)	8	Limits the packet's lifespan
Protocol	8	Defines upper-layer protocol (TCP = 6, UDP = 17)
Header Checksum	16	Ensures integrity of the header
Source Address	32	Sender's IP address
Destination Address	32	Receiver's IP address



This how IP packer looks

Cybersecurity Concerns with IP:

- **IP Spoofing:** Attackers modify the source IP address to disguise their identity.
- **DDoS Attacks:** Attackers flood a network with massive traffic using multiple IPs.
- **Man-in-the-Middle (MITM) Attacks:** Intercepting IP packets to alter or steal data.

Security Measures:

- **IPSec (Internet Protocol Security) –** Encrypts IP packets to ensure data confidentiality.
- **Firewalls & Intrusion Prevention Systems (IPS) –** Filters malicious IP traffic.
- **VPNs (Virtual Private Networks) –** Hides real IP addresses for secure communication.

Transmission Control Protocol (TCP) – Reliable Communication

What is TCP?

TCP (Transmission Control Protocol) is a connection-oriented protocol that ensures **reliable, ordered, and error-checked** delivery of data. It is commonly used for applications requiring **accurate transmission**, such as web browsing, emails, and file transfers.

Key Features of TCP:

- **Connection-Oriented:** Requires a **three-way handshake** before sending data.
- **Reliable Transmission:** Ensures **data integrity** with retransmissions if packets are lost.
- **Flow Control:** Prevents sender from overwhelming receiver.
- **Congestion Control:** Avoids excessive network traffic by adjusting transmission speed.

TCP Three-Way Handshake (Connection Establishment)

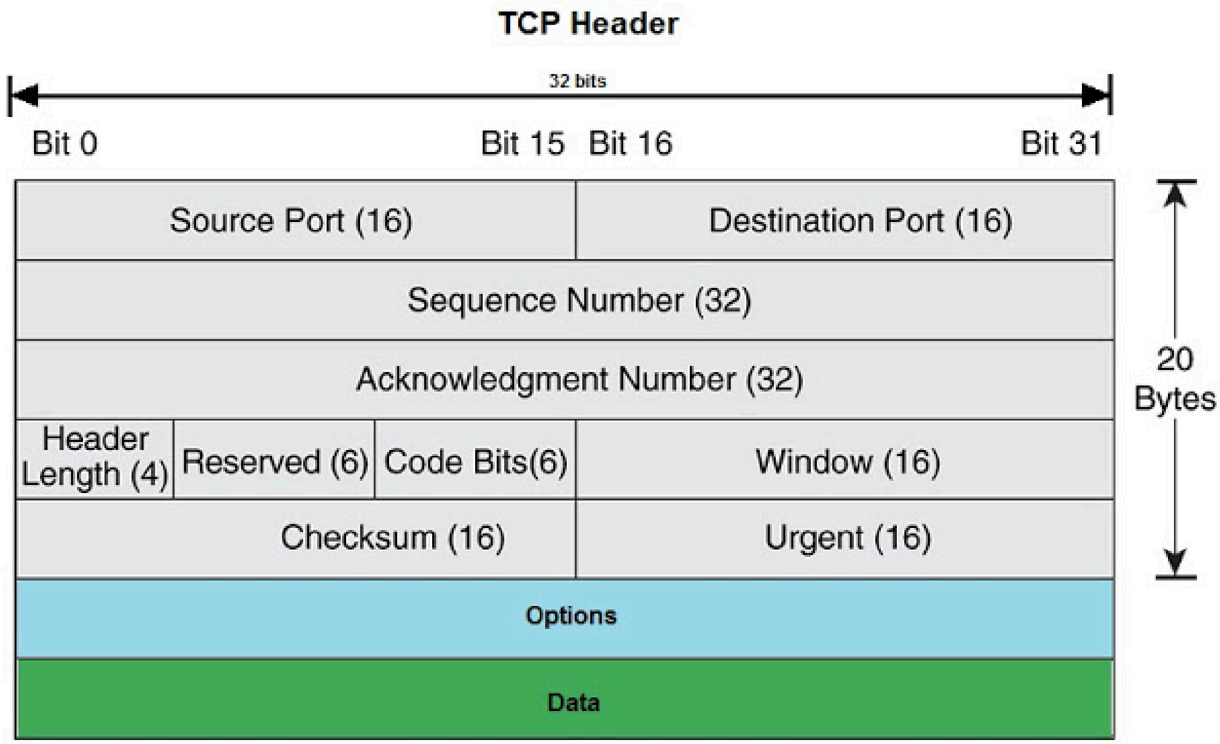
1️⃣**SYN (Synchronize)** – Client sends a SYN request to initiate a connection.

2️⃣**SYN-ACK (Synchronize-Acknowledge)** – Server responds with SYN-ACK.

3️⃣**ACK (Acknowledge)** – Client sends ACK, and connection is established.

TCP Header Fields:

Field	Size (Bits)	Description
Source Port	16	Sender's port number
Destination Port	16	Receiver's port number
Sequence Number	32	Tracks packet order
Acknowledgment Number	32	Confirms received packets
Flags	6	Control bits (SYN,ACK,FIN,etc.)
Window Size	16	Controls flow of data
Checksum	16	Ensures data integrity



Here you can see TCP Header

Cybersecurity Concerns with TCP:

- **TCP SYN Flood Attack:** Sends multiple SYN requests without completing the handshake, leading to **DoS attacks**.
- **Session Hijacking:** Attackers steal session information to impersonate users.
- **MITM Attacks:** Intercepting TCP communication to alter or steal data.

Security Measures:

- ➢ **SYN Cookies** – Prevents SYN Flood Attacks.
- ➢ **TLS/SSL Encryption** – Secures TCP communication.
- ➢ **Firewalls & IDS/IPS** – Detect and block malicious TCP packets.

User Datagram Protocol (UDP) – Fast but Unreliable

What is UDP?

UDP (User Datagram Protocol) is a connectionless protocol that prioritizes **speed** over **reliability**. It is ideal for applications where real-time communication is more important than perfect accuracy, such as video streaming, online gaming, and VoIP calls.

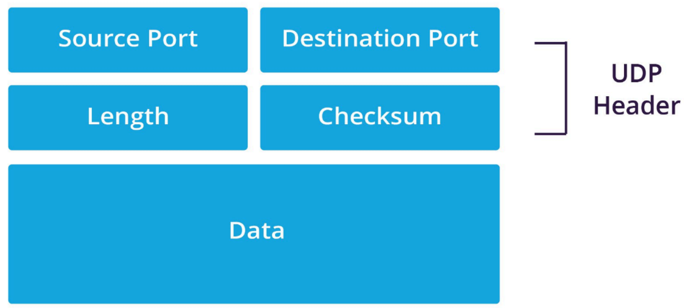
Key Features of UDP:

- **Connectionless:** No handshake, so packets are sent without establishing a connection.
- **Fast Transmission:** No retransmissions or acknowledgments, reducing delay.
- **Low Overhead:** Simpler than TCP, with minimal protocol overhead.

UDP Header Fields:

Field	Size (Bits)	Description
Source Port	16	Sender’s port number
Destination Port	16	Receiver’s port number

Length	16	Size of the UDP packet
Checksum	16	Ensures data integrity



Here you can see UDP Header

Cybersecurity Concerns with UDP:

- **UDP Flood Attack:** Sends massive amounts of UDP traffic to overwhelm a server (**DoS attack**).
- **DNS Amplification Attack:** Uses open DNS resolvers to amplify attack traffic.
- **MITM Attacks:** UDP traffic can be intercepted and modified.

Security Measures:

➢ **Rate Limiting & Firewalls** – Prevent UDP floods.
➢ **DNSSEC (DNS Security Extensions)** – Secures DNS queries.
➢ **Encrypted UDP (DTLS)** – Encrypts UDP traffic for security.

Conclusion

✅ **IP handles addressing and packet routing.**

✅ **TCP ensures reliable, connection-oriented communication**.

✅ **UDP provides fast, connectionless transmission**.

✅ **Cybersecurity threats target all three protocols, requiring proper security measures**.